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324414 (24)

BE (4th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical

DIGITAL ELECTRONICS & LOGIC DESIGN

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question carry 2 marks.

(ii) Attempt any two parts from parts (b), (c) and (d) each carrying 7 marks.

Q. 1. (a) What are alphanumeric codes, where are they used ? 2

(b) (i) Convert decimal number 36.1875 to binary number. 1

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(ii) $(786.983)_{10} = (x)_{16}$, x = ? 2

(iii) $(467.342)_8 = (x)_{10}$, x = ? 2

(iv) $(1A5B)_{16} = (x)_8$ 2

(c) Determine the single error correcting code for BCD number 1001 (information bits) using even parity ? 7

(d) (i) Describe the binary coded decimal (BCD) systems. How will you form a BCD number ? 3

(ii) How will you carry out 2's complement subtraction ? Give the steps of subtraction from smaller number as well as from larger number ? 4

Q. 2. (a) What is the difference between Boolean Algebra and ordinary algebra ? 2

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(b) Simplify using Boolean laws.

(i) $A(B + C\overline{A}) + BC(B + \overline{C}A)$ 1

(ii) $[A\overline{B}C(C + BD) + \overline{A}\,\overline{B}]C$ 2

(iii) Prove that $AB + CD = (A + C)(A + D)(B + C)(B + D)$. 2

(iv) Convert the given expression into canonical SOP form : 2

$f = A + AB + BC$

(c) Simplify the Boolean Expression using k-map $F(A, B, C, D) = \pi M(4, 5, 6, 7, 8, 12) \cdot d(0, 1, 2, 3, 9, 11, 14)$. 7

and implement using universal logic.

(d) Simplify the Boolean function using tabulation method : 7

$F(A, B, C, D) = \Sigma m(0, 2, 3, 6, 7, 8, 10, 12, 13)$

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Q. 3. (a) Implement a 32 × 1 MUX using 16 × 1 and 2 × 1 MUX. 2

(b) Design a combinational circuit whose output is equal to 1 if four input variable have more 1's than 0's. The output is 0 otherwise. 7

(c) Implement the following Boolean function using 8 : 1 MUX :

$F(A, B, C, D) = \Sigma m(1, 2, 4, 7, 11, 13, 15)$ 7

(d) Design a 3 bit odd parity generator and ckecker ? 7

Q. 4. (a) Why are shift registers considered to be basic memory devices ? 2

(b) (i) Show how JK Flip Flop can be operated as toggle Flip Flop ? 3

(ii) Explain how JK Flip Flop can be converted to D-Flip Flop ? 4

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(c) Explain the working of serial in parallel out shift register with logic diagram and pulse waveforms ? 7

(d) Design a Mod-8 synchronous counter ? 7

Q. 5. (a) Draw block diagram of basic-memory operation ? 2

(b) Explain terms :

(i) Memory cell ½

(ii) Memory word ½

(iii) Byte ½

(iv) Capacity 1

(v) Density ½

(vi) Address 1

(vii) Read operation 1

(viii) Write operation 1

(ix) Access time 1

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(c) Give the procedure for analysis and synthesis of sequential circuits with the help of block diagram ? 7

(d) Compare Moore machine and Mealy machine using an eg. 7

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